

Cálculo Diferencial

Asesorías

08-02-26

Ejercicio 1. Encontrar un valor de a tal que $f(a) = \frac{2}{3}$
donde

$$f(x) = \frac{9x+1}{x+2}$$

Sol.

$$\frac{9x+1}{x+2} = \frac{2}{3}$$

$$3(9x+1) = 2(x+2)$$

$$27x+3 = 2x+4$$

$$27x-2x = 4-3$$

$$25x = 1$$

$$x = \frac{1}{25}$$

Comprobar:

$$\frac{9(\frac{1}{25})+1}{\frac{1}{25}+2} = \frac{9+25}{1+50} = \frac{34}{51} = \frac{2}{3}$$

Ejercicio. Graficar

$$f(x) = \begin{cases} \underline{3+x}, & x \in [0,1) \\ \underline{3-x}, & x \in [1,2] \end{cases}$$

solución.

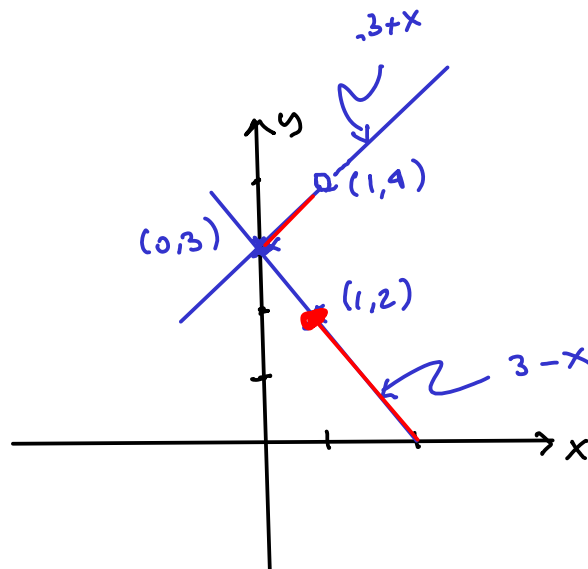
caso 1. $y = \underline{3+x}$

$$\begin{array}{r|l} x & 0 \quad 1 \\ \hline y & 3 \quad 4 \end{array}$$

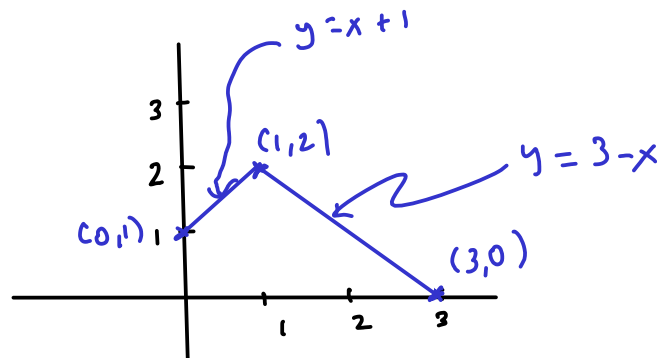
Curso 2

$y = 3 - x$

x	0	1
y	3	2



Ejercicio 3. Encontrar la función que tiene la siguiente gráfica



Solución.

De (0, 1) a (1, 2)

$$m = \frac{2-1}{1-0} = \frac{1}{1} = 1$$

Fórmula Puntos-pendiente

$$y - 1 = m(x - 0)$$

$$y - 1 = 1 \cdot (x)$$

$$y - 1 = x$$

$$y = x + 1$$

De (1, 2) a (3, 0)

$$m = \frac{0-2}{3-1} = \frac{-2}{2} = -1$$

$$y - 2 = -1(x - 1)$$

$$y - 2 = 1 - x, \quad y = 3 - x$$

$$f(x) = \begin{cases} x+1, & x \in [0,1] \\ 3-x, & x \in [1,3] \end{cases}$$